CNC MACHINE OPERATIONS & CARE

CNC Troubleshooting Guide

Machine problems • symptoms • possible causes • safe diagnostic steps

DOCUMENT CONTROL	DETAIL
Document	CNC_Troubleshooting_Guide.pdf
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Applicability	General CNC machining centres, lathes and routers; adapt locally

Important: This is an educational, generic reference. CNC machines differ substantially by builder, model, control, options, tooling and site rules. The machine-specific operator, maintenance, electrical and safety manuals take precedence. Do not use example values, procedures or alarm descriptions as substitutes for approved machine documentation.

1. Purpose and safe diagnostic approach

This guide helps operators and maintenance personnel organize fault-finding by symptom. It is not a substitute for the machine builder's troubleshooting tree, electrical drawings, PLC documentation or approved service procedures.

Use a disciplined sequence:

1. **Make safe.** Stop the cycle. If there is smoke, fire, unusual impact, uncontrolled motion, a serious leak or risk to a person, use the site emergency response procedure and emergency stop as appropriate.
2. **Record the evidence.** Note the exact alarm text/number, time, program and block, axis/spindle, operating mode, tool, work material and what happened immediately before the fault.
3. **Preserve information.** Photograph the display if permitted. Do not clear alarms, reset drives, power-cycle or edit parameters until the fault information has been captured and the approved procedure permits it.
4. **Check simple external causes.** Verify workholding, tool condition, material, doors/interlocks, air/coolant supply and obvious obstructions without entering a hazardous area.
5. **Escalate.** Use qualified maintenance personnel for electrical, hydraulic, pneumatic, servo, spindle, safety-circuit or control-system work.

Never defeat an interlock, bypass a guard, reach into a machine while motion is possible, or repeatedly reset a fault without understanding its cause.

2. Symptom / cause / troubleshooting matrix

Symptom	Possible causes	Checks and response
Machine will not power up	Isolator off; supply interruption; E-stop active; door/safety circuit open; control power fault; blown protection device.	Check site supply status and the documented power-up checklist. Confirm E-stop release and guards only from a safe position. Do not open electrical cabinets or replace fuses unless authorized.
Control powers up but servo/axis not ready	Drive alarm; axis limit; low control voltage; feedback fault; safety chain not satisfied; brake issue.	Capture drive and CNC alarms. Check displayed axis status and permitted external conditions. Follow the builder's drive-reset procedure; qualified personnel investigate feedback, brake and cabinet faults.
Axis following error / position deviation	Collision; excessive acceleration/load; binding; lubrication issue; encoder/scale fault; loose coupling; servo tuning issue.	Stop operation and inspect for visible damage only when isolated and safe. Record axis, direction and commanded/actual position. Do not change servo gains or machine parameters without OEM authorization.
Poor dimensional accuracy	Tool wear; incorrect offset; thermal growth; workpiece movement; backlash; machine not warmed up; wrong compensation or datum.	Verify drawing, datum, tool number, tool length/radius and work offsets. Check clamping and tool wear. Measure a controlled test feature. Maintenance should check geometry, backlash and thermal compensation.
Poor surface finish / chatter	Dull or damaged tool; incorrect speed/feed; tool overhang; weak workholding; runout; resonance; chip recutting.	Stop and inspect tool and setup. Verify holder cleanliness, runout and clamping. Review approved cutting data and chip evacuation. Change one process variable at a time under process-engineering approval.
Spindle will not start / spindle alarm	Door/interlock; spindle drive fault; tool clamp state; lubrication/cooling fault; overload; speed command outside limits.	Record alarm and commanded speed. Check approved interlocks and utility indicators. Do not force spindle enable or alter drive limits. Qualified staff inspect drive, cooling, lubrication and clamp systems.

Symptom	Possible causes	Checks and response
Tool changer stops / tool-change failure	Incorrect pot position; tool not seated; damaged gripper; air pressure low; magazine obstruction; sequence interrupted.	Do not enter the tool changer while it can move. Follow the machine-specific recovery sequence. Check tool/pot identity and visible damage after isolation. Call maintenance for mechanical or sequence faults.
Coolant flow low / no coolant	Low tank level; clogged filter/nozzle; pump fault; blocked line; incorrect coolant concentration; level/flow switch fault.	Stop cutting if coolant is required for safe process control. Check level and visible filter/nozzle condition using approved PPE and isolation. Handle coolant per SDS and site environmental rules.
Hydraulic / pneumatic pressure alarm	Low reservoir; leak; pump/compressor fault; blocked filter; pressure switch fault; temperature issue.	Stop if clamping or safe machine function may be affected. Do not tighten pressurized fittings. Isolate and depressurize using the approved procedure before qualified repair.
Program alarm / unexpected motion	Syntax or modal command issue; wrong work offset; wrong tool call; incorrect plane/units; stale modal state; postprocessor mismatch.	Do not rerun blindly. Review the alarm block and preceding modal commands. Verify units, coordinate system, tool data, work offset and simulation/dry-run procedure. Have the program checked before production.
Excessive vibration / unusual noise	Loose workholding; damaged bearing; tool imbalance; collision damage; worn belt/coupling; chip trapped in enclosure.	Stop the cycle. Do not continue to ‘see if it clears’. Inspect only after safe isolation. Record when noise occurs (axis, speed, load) for maintenance diagnosis.
Chip conveyor jam / overload	Long stringy chips; overfilled bin; foreign object; conveyor overload; coolant/chip buildup.	Stop conveyor and machine as required. Isolate and verify zero energy before clearing a jam. Never pull chips by hand; use designated tools and cut-resistant PPE where specified.

3. Symptom-specific diagnostic notes

Alarm and event capture

- Record the complete alarm wording, number, subcode and timestamp; some controls show multiple related messages.
- Record whether the alarm appears at power-up, homing, tool change, spindle start, cutting load or program end.
- Note recent changes: tooling, fixture, program revision, offsets, maintenance, utility interruption or crash.
- Keep a fault log with corrective action and verification result. A reset that temporarily clears a fault is not proof of repair.

Accuracy and process faults

- Separate machine-positioning error from tool, fixture, material and measurement error.
- Use calibrated measurement equipment and a repeatable measurement method.
- Check thermal state, warm-up routine, tool wear, tool setter results, workholding repeatability and datum strategy.
- After any collision, stop production and request an authorized inspection before returning the machine to service.

4. Return-to-service checklist

- Root cause identified or formally assessed; corrective action completed by an authorized person.
- Guards, doors, interlocks and safety devices restored and tested using the approved method.
- Tools, fixtures, workpiece and loose items secured; enclosure clear of people and obstructions.
- Offsets, tool data, program revision and work coordinate system independently verified.
- A controlled test run completed using the machine builder’s approved prove-out method.
- Fault log updated; operator informed of restrictions or follow-up actions.

5. Troubleshooting record template

Field	Record
Date / time / machine ID	_____
Operator / mode / program & revision	_____
Alarm number and exact text	_____
Axis / spindle / tool / program block	_____
Symptom and conditions when it occurred	_____
Recent changes / previous related events	_____
Checks performed and findings	_____
Corrective action / person responsible	_____
Verification test / result / release approval	_____

6. Escalation triggers

Immediately stop and escalate where there is suspected structural damage, a collision, uncontrolled axis movement, smoke/fire, electrical arcing, repeated drive trips, safety-device failure, hydraulic injection risk, damaged lifting equipment, or uncertainty about stored energy. Only designated competent personnel may diagnose or repair these conditions.